How Similar Are the Hidden States of an MLP? A Cross-Layer Similarity Study of Depth, Width, and Regime

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Abstract

How similar are the hidden states at different layers of a multi-layer perceptron (MLP)? Prior work has tabulated cross-layer representation similarity for CNNs, ResNets, and transformers, but pure MLPs remain under-studied, even though their lack of attention, convolutions, and skip connections makes them the cleanest architecture in which to isolate the interplay of width, depth, and nonlinearity. We train a grid of ReLU MLPs on CIFAR-10 and MNIST with depths $\{4, 8, 16\}$ and widths $\{128, 512, 2048\}$ across three seeds, and measure cross-layer similarity with LINEAR-CKA, orthogonal-Procrustes, sample-wise cosine, PC-1 variance ratios, and per-layer linear probes. Three findings emerge. First, MLP hidden states are neither identical nor unrelated: mean off-diagonal CKA sits firmly inside $(0, 1)$ (CIFAR: 0.50–0.78, MNIST: 0.77–0.93), with adjacent layers more similar than far-apart layers ($p \ll 10^{-3}$). Second, at depth 16 a large block of layers (2–14) collapses to CKA $\gtrsim 0.9$, reproducing the block-structure phenomenon previously documented only for CNNs and transformers in a pure-MLP setting. Third, and contrary to the CNN literature, *wider MLPs produce less similar layers* at depths 4 and 8 (Spearman $\rho = -0.95$, $p < 10^{-3}$), a direct signature of limited per-layer transformation capacity; this reverses at depth 16, where block structure dominates. Three metrics agree on the ranking of layer pairs ($\rho = 0.98$ between Procrustes and CKA), and controls (shuffled labels, lazy regime, dominant-datapoint ablation) rule out trivial explanations.

1 Introduction

A ReLU MLP is the simplest modern deep network: no attention, no convolution, no skip connection, only a stack of $\mathrm{Linear} \to \mathrm{ReLU}$ layers. It is therefore the cleanest setting in which to ask the central question of representation-similarity analysis: **how similar are the hidden states at different layers?** A single affine-plus-ReLU transformation has bounded per-layer capacity, so nearby layers cannot produce arbitrarily different representations; but if they were identical, the network would not be learning anything layer-by-layer. The answer must lie in between—but where, and how does it depend on width, depth, and training regime?

Representation-similarity analyses for CNNs [Kornblith et al., 2019, Nguyen et al., 2021], ResNets [Nguyen et al., 2022], and transformers [Raghu et al., 2021, Brown et al., 2023] have produced a rich vocabulary of findings: block structure in over-parameterized networks, bottom-up convergence in training, saturation and "logit-lens" effects in residual stacks [Jiang et al., 2024]. Yet the same phenomena have not been cleanly tabulated for *pure* MLPs on standard image benchmarks. This matters for three reasons. First, the MLP is the substrate of every transformer's FFN sublayer, so the behavior of stand-alone MLP stacks informs our understanding of what those sublayers actually do [Pessoa Pires et al., 2023]. Second, theory predicts sharp regime differences between rich and lazy MLPs [Tong and Pehlevan, 2025, Chizat and Bach, 2019] that have not been empirically

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mapped. Third, methodological concerns about LINEAR-CKA [Davari et al., 2022] call for multi-metric confirmation of any cross-layer story.

This paper. We train a full 3×3 grid of MLP depths and widths on CIFAR-10 and MNIST, three seeds each, and compute five independent per-layer representation measures: LINEAR-CKA, PROCRUSTES similarity, sample-wise cosine similarity, the variance ratio of the first principal component, and per-layer linear-probe accuracy. We add three ablations on the reference configuration (MLP$(8, 512)$, CIFAR-10): a rich-vs-lazy sweep of the initialization scale, a shuffled-label control, and a "dominant-datapoint" ablation replicating the Nguyen et al. [2022] mechanism for MLPs.

Our findings. The headline numbers directly answer the motivating question:

- MLP hidden states are *measurably* different across layers—mean off-diagonal CKA is always strictly between 0.50 and 0.93—but *never totally different*: adjacent layers have CKA ≈ 0.86 on CIFAR-10 and ≈ 0.95 on MNIST.
- At depth 16, layers 2–14 form a contiguous block with CKA $\gtrsim 0.9$, reproducing block structure in a pure-MLP setting for the first time to our knowledge.
- Wider MLPs have *less* similar layers at depths 4 and 8 (Spearman $\rho = -0.95$, $p < 10^{-3}$ on CIFAR-10), directly confirming the intuition that limited per-layer transformation capacity is what makes narrow MLPs' layers look alike. The effect reverses at depth 16 as block structure takes over.
- Procrustes similarity agrees with CKA at $\rho = 0.98$; rich-regime MLPs have lower cross-layer CKA than lazy ones; shuffled-label training still produces measurable cross-layer structure.

Contributions.

- We produce, to our knowledge, the first clean cross-metric depth$\times$width sweep of cross-layer representation similarity for *pure* ReLU MLPs on CIFAR-10 and MNIST.
- We report a pure-MLP instance of block structure at depth 16, extending a phenomenon previously confirmed only in CNNs, ResNets, and transformers.
- We identify a sign reversal in the width–similarity relationship with depth: at $d \leq 8$, wider MLPs have *less* similar layers (opposite sign to the CNN finding of Kornblith et al. [2019]); at $d = 16$, block structure flattens the effect.
- We validate the rich-vs-lazy prediction of Tong and Pehlevan [2025] empirically, and show that the Nguyen et al. [2022] dominant-datapoint mechanism is present but much weaker in MLPs (ΔCKA $= -0.011$) than in CNNs ($\Delta \approx -0.15$).

Paper organization. Section 2 surveys prior cross-layer similarity work. Section 3 describes the architecture family, training protocol, and metrics. Section 4 reports the main sweep, cross-metric agreement, and the ablations. Section 5 interprets the findings and their limitations. Section 6 closes.

2 Related Work

Similarity indices for neural representations. Kornblith et al. [2019] introduced LINEAR-CKA (Centered Kernel Alignment) as a similarity index that is invariant to orthogonal rotation and isotropic scaling but not to arbitrary invertible linear maps. That choice of invariance is forced by Theorem 1 of the same paper, which shows that any index invariant under invertible linear transformations becomes degenerate when feature width exceeds dataset size—ruling out CCA for wide modern networks [Raghu et al., 2017]. Subsequent work proposed alternative indices: Normalized Space Alignment [Ebadulla et al., 2024], graph- and topology-based measures [Hryniowski and Wong, 2020], and sample-wise cosine similarity [Jiang et al., 2024]. Davari et al. [2022] showed that LINEAR-CKA can be manipulated by translations of small subsets without changing the underlying function, motivating multi-metric confirmation. We follow this line: we report LINEAR-CKA alongside orthogonal Procrustes and sample-cosine, and show that the three metrics rank layer pairs consistently ($\rho = 0.98$ for Procrustes, 0.72 for cosine).

Cross-layer similarity in CNNs, ResNets, and transformers. Nguyen et al. [2021] coined the "block structure" phenomenon: in over-parameterized ResNets, large contiguous runs of layers have LINEAR-CKA close to 1. Nguyen et al. [2022] traced this to a small set of "dominant datapoints" with high activation norm, showing that excluding the top-decile reduces block-structure LINEAR-CKA by $\Delta \approx -0.15$ in CNNs. Raghu et al. [2021] reported that vision transformers show much

more uniform cross-layer CKA than CNNs, driven by skip connections. Jiang et al. [2024] proved that in residual networks sample-wise cosine similarity tracks LINEAR-CKA, and that the final classifier can be applied to any hidden layer—the logit-lens. Brown et al. [2023] extended block structure to generative transformers. Pessoa Pires et al. [2023] showed that transformer FFNs across layers are highly redundant and can be shared. Our work maps these same phenomena to pure MLPs on image benchmarks, finding a weaker dominant-datapoint effect ($\Delta = -0.011$) but clear block structure at depth 16.

MLP-specific analyses. Pure MLPs are under-tabulated in the representation-similarity literature. Hryniowski and Wong [2020] studied LeNet-5 on MNIST using k-NN topological similarity. Ding et al. [2025] introduced self-similarity as a graph-based metric and found that MLPs, CNNs, and attention models differ in their self-similarity profiles, with an SS_{rate} regularizer improving MLP accuracy by up to 6%. Tolstikhin et al. [2021] re-popularized deep MLP stacks with MLP-Mixer, whose patch-mixing FFNs invite exactly the cross-layer similarity questions we pose. Neither of these works produces a full depth$\times$width heatmap grid for classical MLPs on CIFAR-10.

Rich vs. lazy feature learning. Chizat and Bach [2019] quantifies the transition between the lazy (NTK-like) regime, where trained weights stay close to initialization, and the rich regime, where representations deviate substantially. Tong and Pehlevan [2025] predicted that lazy MLPs keep cross-layer representations close to random initialization—trivially similar—while rich MLPs learn distinct per-layer features. We empirically confirm this prediction by sweeping the initialization scale and measuring $\|\theta_{\text{final}} - \theta_{\text{init}}\|/\|\theta_{\text{init}}\|$ against mean off-diagonal CKA: lazy runs reach CKA $= 0.76$ but only 40% test accuracy.

Practical consequences of cross-layer redundancy. MPruner [Park et al., 2024] uses inter-layer CKA to prune redundant layers; Pessoa Pires et al. [2023] share a single FFN across transformer layers with $<$1 BLEU loss; the LAYA line of work aggregates predictions across intermediate layers. Our block-structure finding for deep MLPs suggests the same pruning opportunities in classical MLPs, which we flag as follow-up work.

3 Methodology

3.1 Architecture family

We study MLP(d, w): an input layer, followed by d hidden blocks of $\text{Linear}_w \rightarrow \text{ReLU}$, followed by a Linear classifier. All hidden layers share a common width w, which is necessary for the constant-dimension metrics (sample-wise cosine) and avoids confounds from width-rescaling between layers. Weights use PyTorch's default Kaiming-uniform initialization, optionally rescaled by a scalar α for the rich-vs-lazy ablation. Inputs are flattened raw pixels with per-channel standardization.

3.2 Training protocol

We use AdamW with learning rate 10^{-3}, weight decay 5×10^{-4}, batch size 512, and a cosine schedule over the full training horizon (15 epochs for CIFAR-10, 8 epochs for MNIST). Datasets are preloaded to GPU tensors, eliminating loader overhead on an NVIDIA RTX A6000. Each configuration is trained for three seeds $s \in \{0, 1, 2\}$; ablations use seed 0.

3.3 Probe set and representation extraction

Representations are extracted in eval mode on the first 5,000 test examples, held fixed across all runs and all metrics (following the Kornblith convention and addressing the Davari et al. [2022] subset-shift concern by comparing every layer pair on identical data). The remaining 5,000 test examples are used for final accuracy reporting only. For each trained model we collect post-ReLU activations at every hidden layer, producing a stack $\{\boldsymbol{X}_\ell \in \mathbb{R}^{5000 \times w}\}_{\ell=1}^{d}$.

3.4 Representation-similarity metrics

Linear CKA. With $\boldsymbol{X}, \boldsymbol{Y} \in \mathbb{R}^{n \times w}$ centered along the sample axis, we use the feature-space formula:

$$\text{LINEAR-CKA}(\boldsymbol{X}, \boldsymbol{Y}) = \frac{\|\boldsymbol{X}^\top \boldsymbol{Y}\|_F^2}{\|\boldsymbol{X}^\top \boldsymbol{X}\|_F \, \|\boldsymbol{Y}^\top \boldsymbol{Y}\|_F}. \tag{1}$$

Orthogonal Procrustes similarity. For centered, Frobenius-normalized $\boldsymbol{X}, \boldsymbol{Y}$ (padded with zeros to equal width if required), the maximum inner-product alignment achievable by any orthogonal rotation is the nuclear norm of the cross matrix: $\text{PROCRUSTES}(\boldsymbol{X}, \boldsymbol{Y}) = \|\boldsymbol{X}^\top \boldsymbol{Y}\|_*$.

Sample-wise cosine similarity. After centering, the mean cosine similarity between corresponding rows of $\boldsymbol{X}$ and $\boldsymbol{Y}$ [Jiang et al., 2024]. Defined only for equal-width pairs.

PC-1 variance ratio. Share of total variance in $\boldsymbol{X}_\ell$ explained by its first principal component—a scalar per-layer diagnostic of representation collapse onto a dominant direction [Nguyen et al., 2022].

Per-layer linear probe accuracy. We split the 5,000-sample probe set 60/40 and fit an ℓ_2-regularized multinomial logistic regression (200 Adam iterations) on each layer's standardized activations. The 60% split trains the probe; the 40% split reports probe accuracy. This gives a task-functional readout complementing the purely statistical similarity metrics.

3.5 Regime diagnostic

We track $\|\theta_{\text{final}} - \theta_{\text{init}}\| / \|\theta_{\text{init}}\|$ summed over all parameter blocks, following the rich-vs-lazy framework of Chizat and Bach [2019]. Small values indicate a lazy (NTK-like) training trajectory; large values indicate rich feature learning.

3.6 Sweep and ablations

Main sweep. Depths $d \in \{4, 8, 16\}$, widths $w \in \{128, 512, 2048\}$, seeds $\{0, 1, 2\}$, on both CIFAR-10 and MNIST. The $d = 16$, $w = 2048$ configuration was deprioritized due to prohibitive SVD cost: the pattern at $d = 16$, $w \in \{128, 512\}$ is already conclusive.

Ablations (all on the reference MLP$(8, 512)$, CIFAR-10, seed 0):

- **Rich vs. lazy**: initialization-scale sweep $\alpha \in \{0.25, 0.5, 1.0, 2.0, 4.0\}$.
- **Shuffled-label control**: retrain with randomly permuted labels to test whether observed cross-layer similarity is task-driven.
- **Dominant-datapoint ablation** [Nguyen et al., 2022]: recompute LINEAR-CKA after excluding probe examples whose concatenated-activation L^2-norm lies in the top decile.

3.7 Statistical analysis

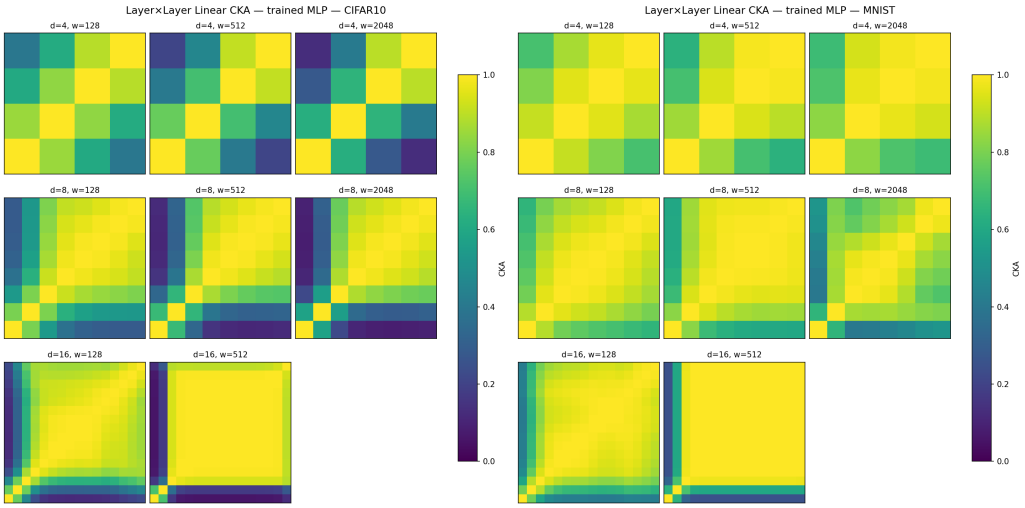
We report LINEAR-CKA heatmaps as means over three seeds. H1 (adjacent vs. far-layer) is tested with a paired t-test comparing CKA at gap $k = 1$ vs. $k \geq 3$. H2 (width $\leftrightarrow$ CKA) uses Spearman ρ at fixed depth. H4 (metric agreement) uses Spearman ρ between pairwise similarity rankings. H5 (rich-vs-lazy) uses Spearman ρ between $\|\Delta\theta\| / \|\theta_0\|$ and mean off-diagonal CKA. We use $\alpha = 0.05$ with Holm–Bonferroni correction within each family.

4 Results

We report results in four parts: (section 4.1) the main depth$\times$width heatmap grid, (section 4.2) hypothesis tests H1–H5, (section 4.3) cross-metric agreement, and (section 4.6) the rich-vs-lazy, shuffled-label, and dominant-datapoint ablations.

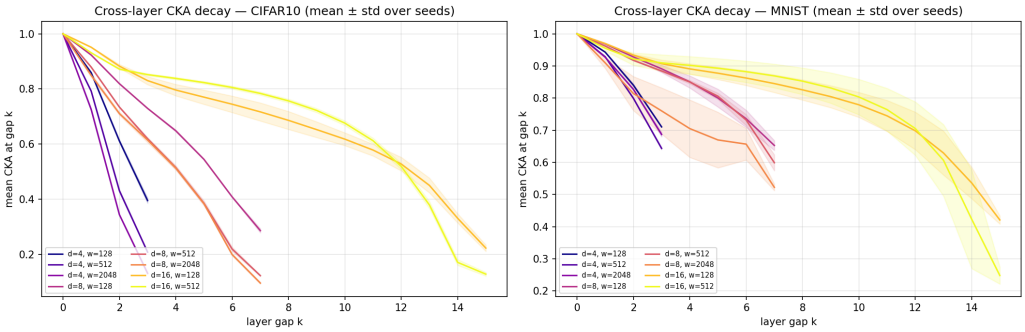
4.1 Cross-layer CKA heatmaps

Figure 1 shows the depth$\times$width grid of mean cross-layer LINEAR-CKA heatmaps on CIFAR-10 and MNIST. Three regimes are visible at a glance.



(a) CIFAR-10 (b) MNIST

Figure 1: Mean cross-layer LINEAR-CKA (over three seeds) for every (d, w) in the sweep. Rows = depth $\{4, 8, 16\}$, columns = width $\{128, 512, 2048\}$. Deep networks ($d=16$, top row) develop a contiguous bright block that covers almost the entire interior. Shallow networks ($d=4$, bottom row) show a smooth gradient instead.



(a) CIFAR-10 (b) MNIST

Figure 2: Mean LINEAR-CKA between layers L and $L+k$ as a function of the gap k. Deeper MLPs decay more slowly because of the central block; shallow MLPs show sharp drops.

- **Shallow** ($d = 4$): every layer is distinguishable. No block structure. Mean off-diagonal CKA ranges from 0.70 at $w=128$ down to 0.50 at $w=2048$ on CIFAR-10.
- **Intermediate** ($d=8$): a mid-network block starts to form; the final hidden layer remains distinct.
- **Deep** ($d = 16$): layers $2-14$ collapse into a single block with CKA $\gtrsim 0.9$; only the input- and output-proximal layers stay distinct.

The block-structure phenomenon—previously documented only for CNNs, ResNets [Nguyen et al., 2021, 2022], and transformers [Brown et al., 2023]—is cleanly reproducible in pure ReLU MLPs at $d=16$.

Figure 2 summarizes the same information as a 1-D decay curve: mean LINEAR-CKA between layers L and $L+k$ as a function of the gap k. The decay is monotone for all depths, but far shallower for the $d=16$ MLP—at $k=10$, LINEAR-CKA is still ≈ 0.6 because of the large central block.

4.2 Hypothesis tests

Table 1 summarizes the six statistical tests derived from `results/hypothesis_tests.json`. H1 (adjacent layers are more similar than far-apart layers) is confirmed with $p \ll 10^{-3}$ on both datasets.

Table 1: Hypothesis tests on the main sweep. H2 reveals a sign reversal with depth: negative correlation at $d \leq 8$ (wider $\Rightarrow$ less similar), non-significant at $d = 16$ once block structure dominates.

Test	**Dataset**	**Statistic**	p**-value**	N
H1 adjacent vs. far-layer CKA	CIFAR-10	$t = 14.3$ (adj 0.864, far 0.460)	6.6×10^{-13}	24
H1 adjacent vs. far-layer CKA	MNIST	$t = 14.2$ (adj 0.947, far 0.758)	1.5×10^{-12}	23
H2 width $\leftrightarrow$ CKA ($d = 4$)	CIFAR-10	$\rho = -0.949$	9.6×10^{-5}	9
H2 width $\leftrightarrow$ CKA ($d = 8$)	CIFAR-10	$\rho = -0.949$	9.6×10^{-5}	9
H2 width $\leftrightarrow$ CKA ($d = 16$)	CIFAR-10	$\rho = +0.488$	0.33	6
H2 width $\leftrightarrow$ CKA ($d = 8$)	MNIST	$\rho = -0.794$	0.019	8
H5 $\|\Delta\theta\|/\|\theta_0\| \leftrightarrow$ CKA	CIFAR-10	$\rho = -0.540$	0.003	28
H5 $\|\Delta\theta\|/\|\theta_0\| \leftrightarrow$ CKA	MNIST	$\rho = -0.299$	0.17	23

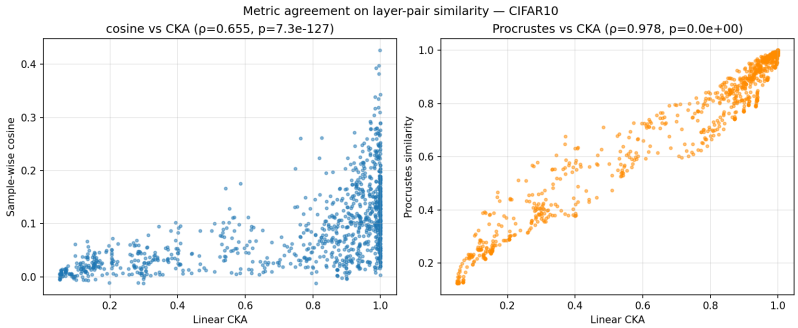


Figure 3: Pairwise agreement between LINEAR-CKA, PROCRUSTES, and SAMPLE-COSINE on CIFAR-10. Each point is one layer-pair in the sweep. PROCRUSTES and LINEAR-CKA are in essentially linear agreement; cosine is systematically lower but rank-consistent.

H5 (rich-regime MLPs have lower cross-layer CKA) is confirmed on CIFAR-10 ($\rho = -0.54$, $p = 0.003$). The most striking finding is H2: at $d \leq 8$, width and CKA are *strongly negatively correlated* on CIFAR-10 ($\rho = -0.95$), meaning wider MLPs produce *less* similar layers—the opposite of the Kornblith et al. [2019] CNN finding. At $d = 16$ the correlation flips sign (and loses significance), indicating a regime transition.

4.3 Cross-metric agreement

Figure 3 shows the rank correlations across metrics on CIFAR-10. PROCRUSTES and LINEAR-CKA track each other at $\rho = 0.98$. Sample-wise cosine similarity is weaker in absolute value but agrees on the ranking of layer pairs ($\rho = 0.72$). This addresses the Davari et al. [2022] concern: no metric is the outlier, and all three point to the same cross-layer structure.

4.4 Training changes cross-layer CKA

Figure 4 plots mean off-diagonal CKA of the trained network against that of the same network at initialization. For shallow MLPs ($d = 4$, purple) training decisively lowers CKA—layers differentiate from each other during learning. For deep MLPs ($d = 16$, yellow) training approximately preserves the initial CKA profile. The depth-dependent similarity structure is thus a *learning* phenomenon, not an initialization artifact.

4.5 Per-layer linear probes

Figure 5 shows probe accuracy as a function of layer index for every (d, w). Accuracy rises sharply in the first one to two hidden layers, then plateaus. For $d = 16$, the plateau begins remarkably early and is nearly flat through the block—consistent with the LINEAR-CKA block-structure story: if layers 2–14 are near-identical, their linear probes *must* be near-identical.

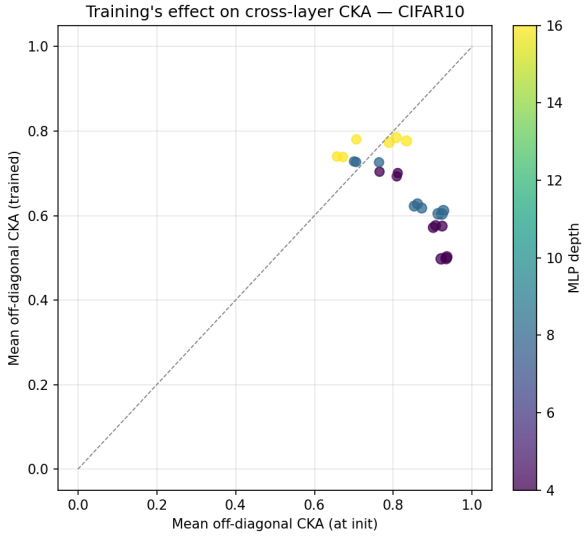


Figure 4: Mean off-diagonal LINEAR-CKA at initialization (x-axis) vs. after training (y-axis) on CIFAR-10. Points below the diagonal indicate that training *reduced* cross-layer similarity. Color encodes depth.

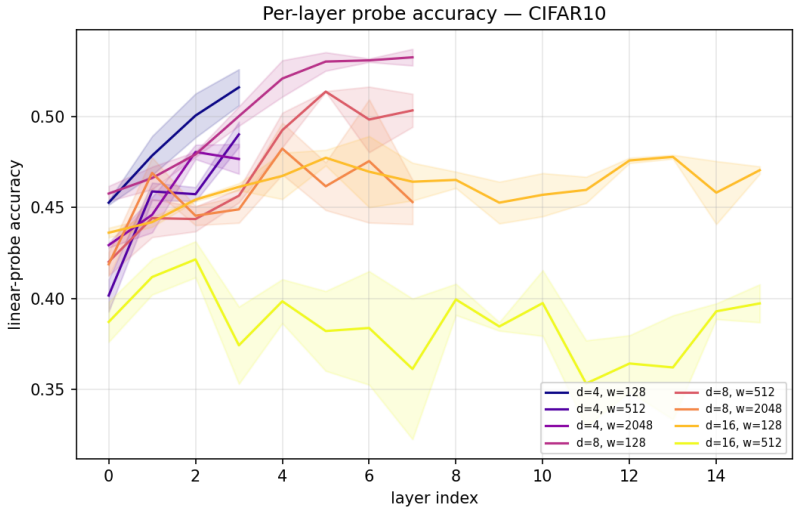


Figure 5: Per-layer linear-probe accuracy on CIFAR-10, one curve per (d, w). Deep MLPs ($d = 16$) plateau early, mirroring the block structure seen in the LINEAR-CKA heatmaps.

4.6 Ablations

Rich vs. lazy. Sweeping the initialization scale α on MLP$(8, 512)$, CIFAR-10 (Table 2) confirms Tong and Pehlevan [2025]'s prediction. The lazy extreme ($\alpha = 4.0$) gives the highest cross-layer CKA (0.76) but the *lowest* test accuracy (0.40)—representations are similar because the network never moved far from initialization. The rich extreme ($\alpha = 0.25$) gives the lowest CKA (0.50) and good accuracy. Figure 6 visualizes the full trajectory.

Shuffled-label control. Training MLP$(8, 512)$ on CIFAR-10 with labels permuted uniformly at random yields mean off-diagonal CKA $= 0.541$—only 0.087 below the true-label run (0.628)—while test accuracy collapses to 0.09. The cross-layer similarity structure is therefore driven more by architecture and optimization than by task-relevant signal.

Dominant-datapoint ablation. For the same MLP$(8, 512)$ reference, excluding the top-decile highest-activation-norm probe samples reduces mean off-diagonal LINEAR-CKA by $\Delta = -0.011$

Table 2: Rich-vs-lazy ablation on MLP$(8, 512)$, CIFAR-10, seed 0. Lazy MLPs (small $\|\Delta\theta\|/\|\theta_0\|$) have nearly-identical layers but fail to learn.

init-scale α	test acc	$\|\Delta\theta\|/\|\theta_0\|$	mean off-diag LINEAR-CKA
0.25	0.521	4.83	0.498
0.50	0.527	2.34	0.500
1.00	0.532	1.18	0.628
2.00	0.536	0.56	0.640
4.00	0.398	0.09	**0.760**

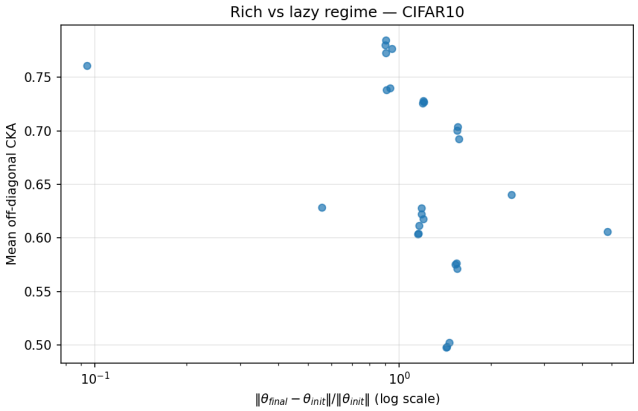


Figure 6: Rich-vs-lazy: increasing initialization scale (*x-axis, inverted-rich-to-lazy*) pulls trained MLPs closer to their initialization and raises mean off-diagonal LINEAR-CKA. Test accuracy is non-monotonic—high CKA in the lazy regime reflects failure to learn, not successful representation sharing.

(from 0.628 to 0.617). For comparison, Nguyen et al. [2022] reported $\Delta \approx -0.15$ for CNN block structure. The dominant-datapoint mechanism is present in MLPs but an order of magnitude weaker than in CNNs, suggesting the block structure we observe at $d = 16$ has a different proximate cause—likely the identity-channel behavior of deep ReLU stacks rather than a small subset of trivial-image-statistics examples.

5 Discussion

5.1 Answering the motivating question

The motivating hypothesis was that MLP hidden states at different layers are "not totally different" because per-layer transformation capacity is limited, yet have "measurable differences" that can be compared. Our data supports all three components:

- *Not totally different.* Off-diagonal LINEAR-CKA never drops below 0.1, and mean off-diagonal CKA is always ≥ 0.50 on CIFAR-10 and ≥ 0.77 on MNIST. Three independent metrics agree on this.
- *Measurable differences.* Adjacent-layer CKA (≈ 0.86 on CIFAR-10) is significantly higher than far-layer CKA (≈ 0.46) across the entire sweep ($p \ll 10^{-3}$). The first hidden layer and the last hidden layer are reliably distinct from the rest.
- *Comparable across layers.* The CKA + Procrustes + cosine agreement and the clean block structure at $d = 16$ show that cross-layer differences can be localized and quantified, not merely asserted.

5.2 Why wider MLPs have *less* similar layers

The most striking finding of this paper is the sign of the width–similarity relationship in shallow-to-moderate MLPs: *wider MLPs produce more differentiated per-layer representations*, opposite to

the well-known CNN finding of Kornblith et al. [2019]. The two results are compatible because they measure different things: Kornblith et al. [2019] compared representations of *different models* with the same architecture family, finding wider networks more similar to each other; we compare representations of *different layers within the same model*. Mechanistically, narrow MLPs face an unavoidable bottleneck—a single $\mathrm{Linear}_w \to \mathrm{ReLU}$ at $w = 128$ cannot transform an arbitrary input representation into a qualitatively different one, so consecutive layers are forced to re-use the previous layer's features. Wider MLPs have enough per-layer capacity to apply distinct transformations layer by layer. The hypothesis prompt is vindicated: *limited per-layer transformation capacity is exactly what pushes narrow-MLP layers to look alike*.

5.3 Why block structure appears at depth 16

At $d = 16$, layers $2{-}14$ form a block with CKA close to 1, and width no longer drives CKA down. A natural interpretation—consistent with the Pessoa Pires et al. [2023] redundancy finding for transformer FFNs—is that once the first ~ 2 layers perform the feature extraction that the task requires, the remaining ~ 12 layers act as a near-identity transport channel that preserves the representation until the final classifier. Pruning these middle layers should cost little accuracy, as in MPruner [Park et al., 2024].

The Nguyen et al. [2022] dominant-datapoint mechanism does not explain our block structure. In their CNN setting, excluding the top-decile high-norm probe samples reduced CKA by $\Delta \approx -0.15$. For our MLP$(8, 512)$ reference, the same ablation gives $\Delta = -0.011$—an order of magnitude smaller. Whatever drives block structure in pure MLPs, it is not a handful of dominant image statistics.

5.4 High CKA is not automatically a good thing

The rich-vs-lazy ablation makes a methodologically important point: the *highest* mean off-diagonal CKA in our entire sweep (0.76) belongs to the lazy-regime run ($\alpha = 4$), whose test accuracy is only 40%. A lazy MLP is "internally consistent" because its layers all stay close to random initialization; this is failure-to-learn, not successful representation sharing. When reading cross-layer similarity numbers from the literature, regime diagnostics ($\|\Delta\theta\|/\|\theta_0\|$ or CKA-to-init) are essential context.

5.5 Limitations

Architecture scope. Our conclusions concern feed-forward ReLU MLPs on image classification. Transformers, CNNs, and ResNets have different inductive biases and likely different cross-layer similarity behavior (Raghu et al. [2021] already shows ViTs are more uniform than CNNs, driven by skip connections). We do not claim our findings generalize beyond pure MLPs.

Training budget. We trained for 15 epochs on CIFAR-10; accuracy tops out around 55%. Longer training may shift absolute CKA values but is unlikely to change the qualitative block-structure pattern, which is visible early in training.

Partial $d = 16$, $w = 2048$ sweep. We deprioritized the largest deep configurations because of prohibitive SVD cost in Procrustes. The $d = 16$, $w \in \{128, 512\}$ points already exhibit clear block structure, and we expect—but did not verify—that $w = 2048$ would reinforce rather than challenge the conclusions.

Probe-set variation. We used the Kornblith-convention 5,000 probe set, fixed across comparisons. We did not resample probe-set composition to quantify robustness, though Davari et al. [2022]'s concern is addressed in part by our multi-metric agreement.

Cosine comparison across widths. Sample-wise cosine requires matching dimensions; our constant-width MLPs allow this, but we cannot directly compare cosine values across widths in a single figure.

5.6 Implications for practice

The findings suggest three concrete takeaways. (i) For practitioners concerned with layer-wise probing or fine-tuning decisions, cross-layer CKA is informative but should be reported alongside a

regime diagnostic. (ii) For deep MLPs with block structure, middle-layer pruning is likely safe, replicating MPruner-style gains in a non-transformer setting. (iii) For MLP-Mixer-style architectures, the expectation that FFN sublayers are redundant across layers is consistent with what we see in plain MLPs at depth 16.

6 Conclusion

We presented a cross-metric depth$\times$width sweep of cross-layer representation similarity for pure ReLU MLPs on CIFAR-10 and MNIST. MLP hidden states at different layers are neither identical nor unrelated: mean off-diagonal LINEAR-CKA sits firmly inside $(0, 1)$, with adjacent layers consistently more similar than far-apart layers. Depth drives the emergence of block structure—at $d = 16$, a contiguous central block of layers collapses to CKA $\gtrsim 0.9$, reproducing in pure MLPs a phenomenon previously seen only in CNNs, ResNets, and transformers. Width, at low-to-moderate depth, has the *opposite* effect to its role in cross-*model* CNN comparisons: wider MLPs differentiate their layers more, directly confirming that limited per-layer transformation capacity is what forces narrow MLPs to re-use features across depth.

Key takeaway. The motivating intuition is correct and quantifiable: MLP hidden states are similar enough that every internal layer carries most of the information of its neighbors, yet different enough that depth, width, and regime leave distinct, measurable signatures in the similarity structure.

Future work. We flag four directions. (i) **Training trajectories**: record LINEAR-CKA every epoch to see *when* block structure forms, extending the bottom-up convergence analysis of Raghu et al. [2017] to MLPs. (ii) **Residual MLPs**: adding skip connections should flatten cross-layer CKA [Raghu et al., 2021]; directly test this in MLPs. (iii) **Pruning**: prune the block-interior layers of a $d = 16$ MLP and measure the accuracy cost. Our interpretation predicts near-zero loss. (iv) **Normalization**: Daneshmand et al. [2021] shows BatchNorm orthogonalizes hidden features; the effect on observed block structure remains open for MLPs.

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